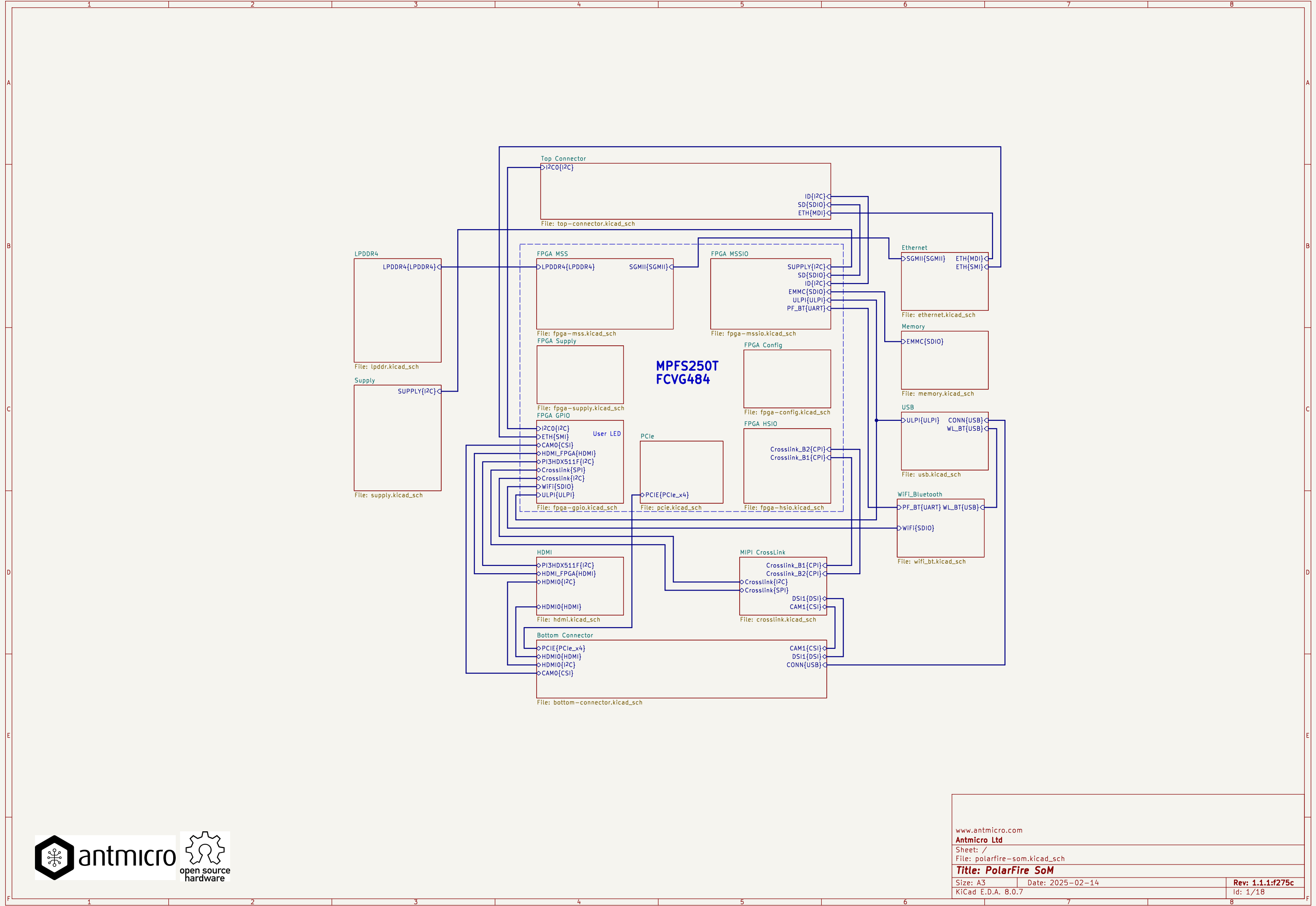
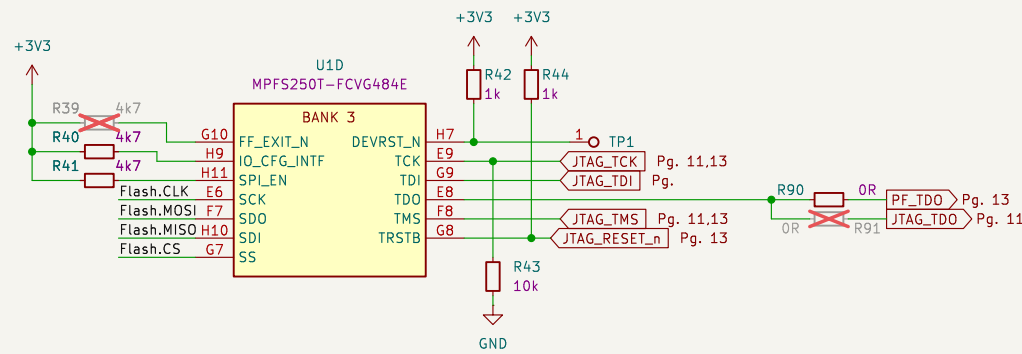
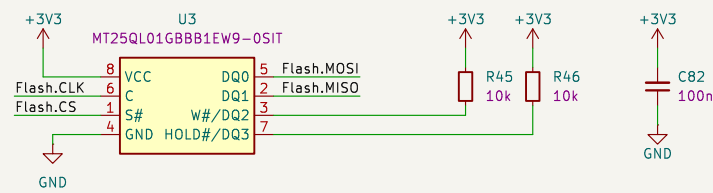




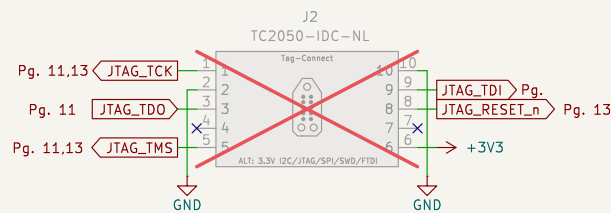
PolarFire Bank 3
JTAG/DEDIO



SPI FLASH



JTAG



The schematic diagram illustrates the power supply circuit for the PolarFire device (U1: MPFS250T-FCVG484E). The device is shown with its various power and signal pins connected to external components. The power pins are connected to a network of voltage sources and ground. The signal pins are connected to a 10k resistor (R14) and ground. The circuit is labeled with 'Pg. 4' on both sides.

Power Pins and Connections:

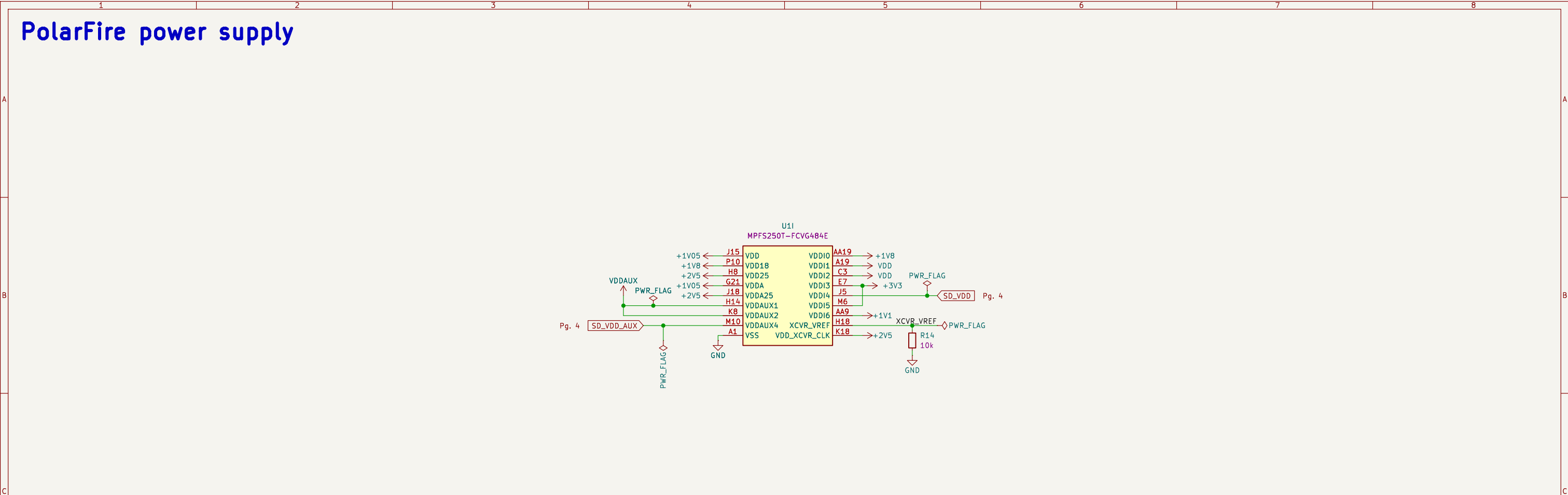
- VDD:** Connected to +1V05 (J15).
- VDD18:** Connected to +1V8 (P10).
- VDD25:** Connected to +2V5 (H8).
- VDDA:** Connected to +1V05 (J18).
- VDDA25:** Connected to +2V5 (H14).
- VDDAUX1:** Connected to +1V05 (J18).
- VDDAUX2:** Connected to +2V5 (H14).
- VDDAUX4:** Connected to +1V1 (AA9).
- VSS:** Connected to GND (A1).

Signal Pins and Connections:

- XCVR_VREF:** Connected to +1V1 (AA9).
- VDD_XCVR_CLK:** Connected to +2V5 (K18) through a 10k resistor (R14) to GND.

Other Components and Connections:

- PWR_FLAG:** Connected to VDDAUX and VDDAUX4.
- SD_VDD_AUX:** Connected to VDDAUX.
- SD_VDD:** Connected to VDDAUX4.
- R14:** 10k resistor connected to VDD_XCVR_CLK and GND.

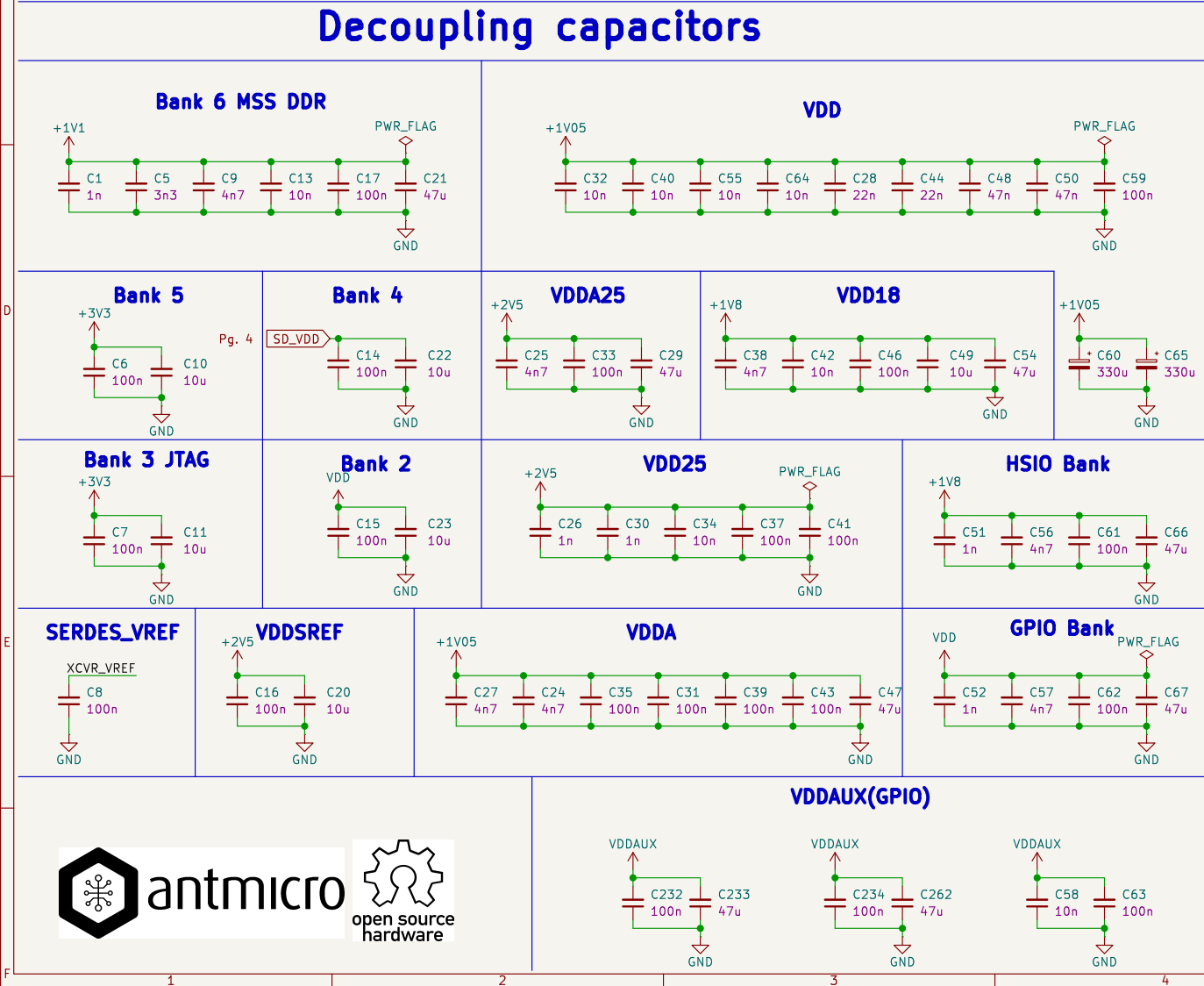
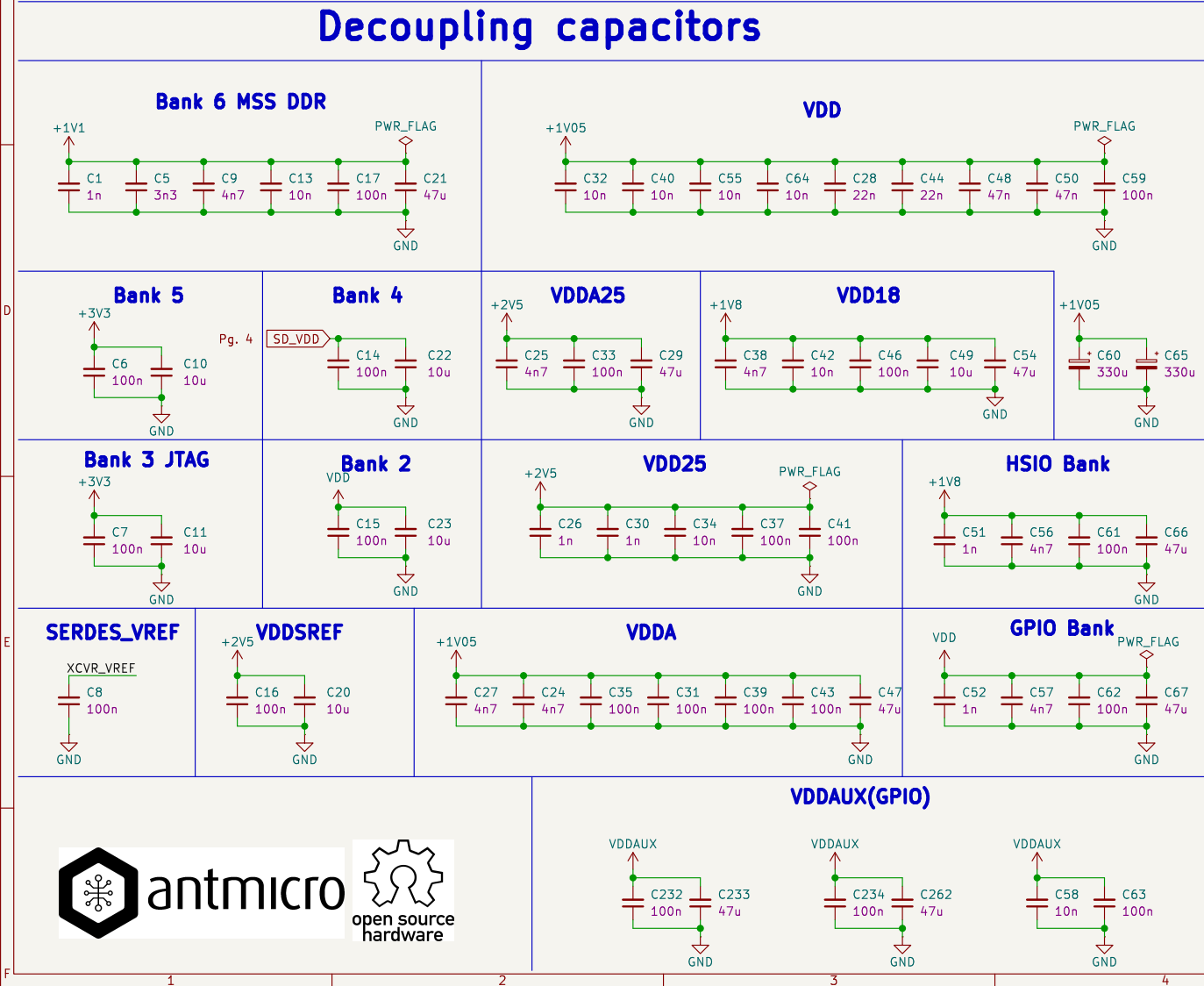


Decoupling capacitors

Bank 6 MSS DDR 		VDD 	
Bank 5 	Bank 4 	VDDA25 	VDD18
Bank 3 JTAG 	Bank 2 	VDD25 	
SERDES_VREF 		VDDAUX(GPIO) 	
VDSREF 		HSIO Bank 	
GPIO Bank 		VDDAUX(GPIO) 	

antmicro
open source hardware

1
2
3
4



www.antmicro.com
Antmicro Ltd
Sheet: /FPGA Supply/
File: fpga-supply.kicad_sch
Title: PolarFire SoM

Size: A3	Date: 2024-06-10	Rev: 1.1.0:f275c
KiCad E.D.A. 8.0.7		Id: 3/18

eMMC

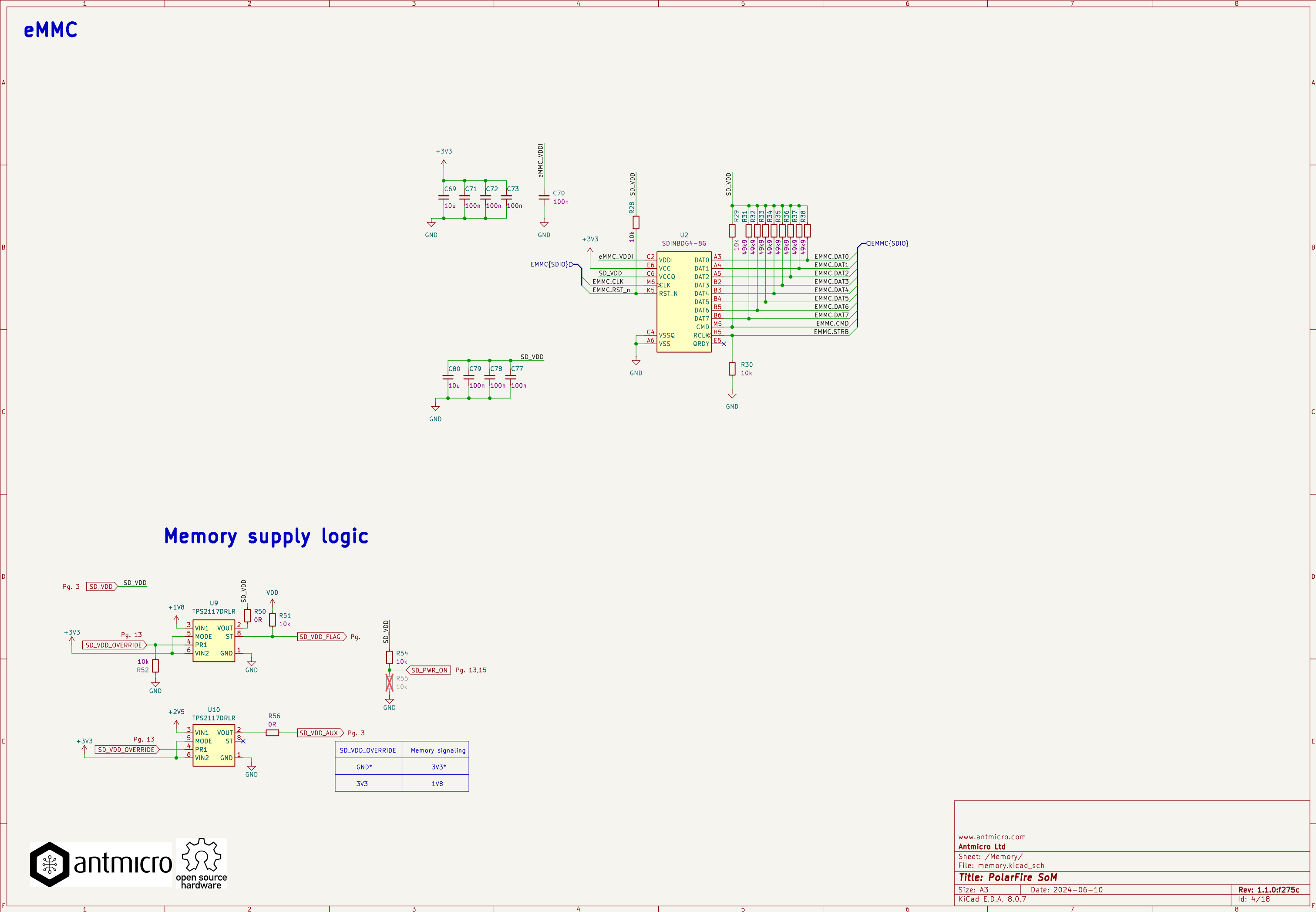
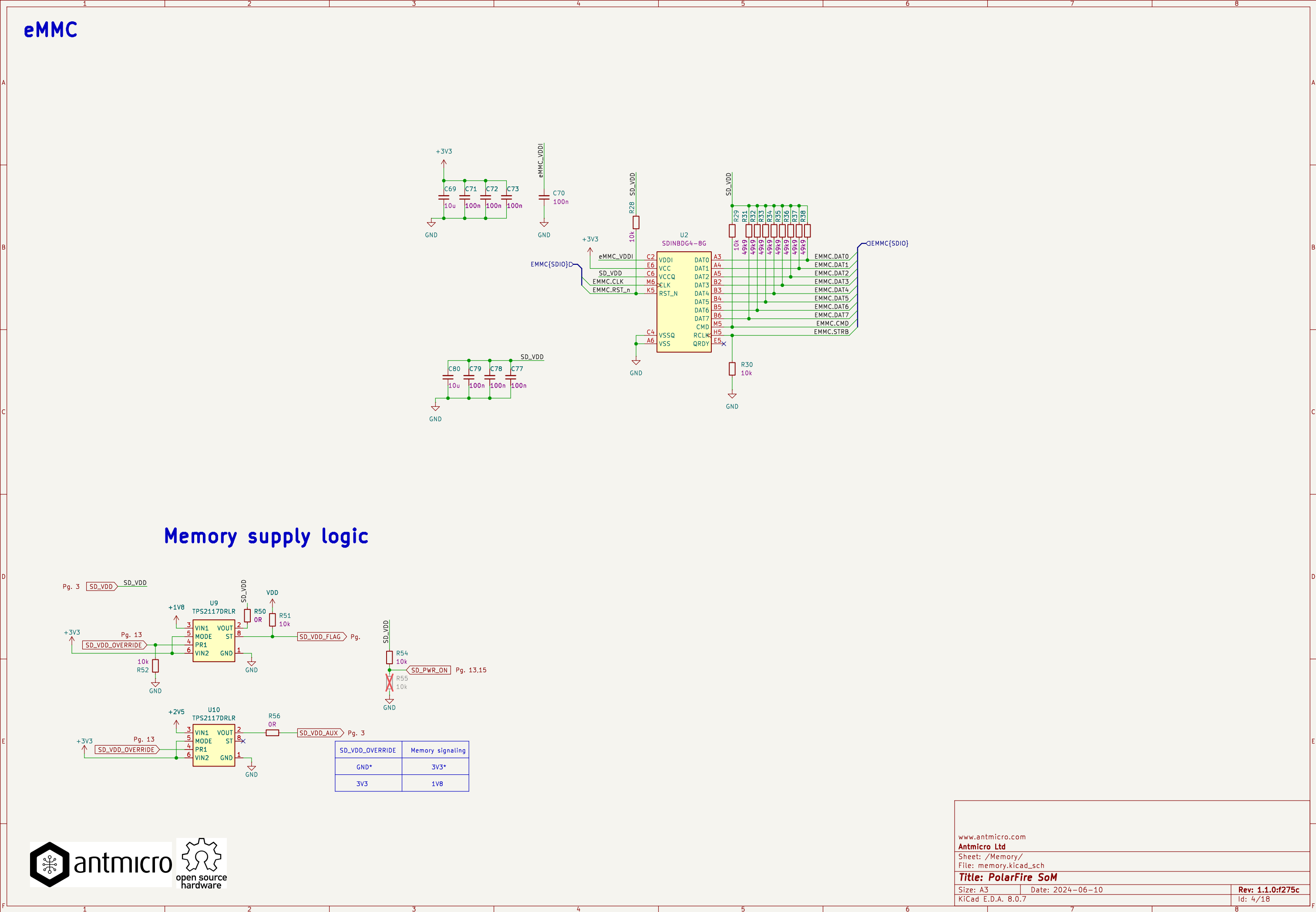
The schematic illustrates the power and signal connections for the eMMC and SD cards. Key components include:

- Power Regulation:** Two TPS2117DRLR voltage regulators (U9 and U10) are used to regulate the SD_VDD_OVERRIDE signal to +1V8 and +2V5, respectively. These are connected to the SD_VDD and SD_VDD_AUX pins of the SD card.
- Decoupling:** Multiple decoupling capacitors (C69, C71, C72, C73, C70, C80, C79, C78, C77) are placed at various power supply nodes to filter noise.
- Signal Connections:** The eMMC is connected to the SDINBDG4-8G (U2) via a 10k resistor (R28). The SD card is connected to the SD_VDD and SD_VDD_AUX pins via 10k resistors (R54, R55).
- Table:** A table defines the SD_VDD_OVERRIDE signal levels for memory signaling.

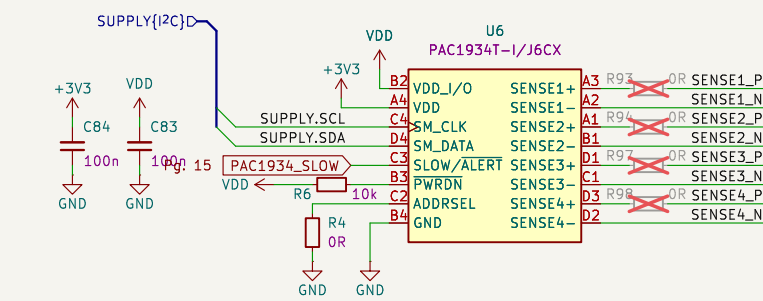
SD_VDD_OVERRIDE	Memory signaling
GND*	3V3*
3V3	1V8

www.antmicro.com
Antmicro Ltd
 Sheet: /Memory/
 File: memory.kicad_sch
Title: PolarFire SoM

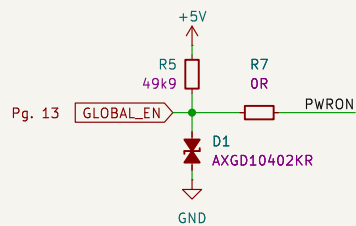
Size: A3	Date: 2024-06-10	Rev: 1.1.0:f275c
KiCad E.D.A. 8.0.7		Id: 4/18



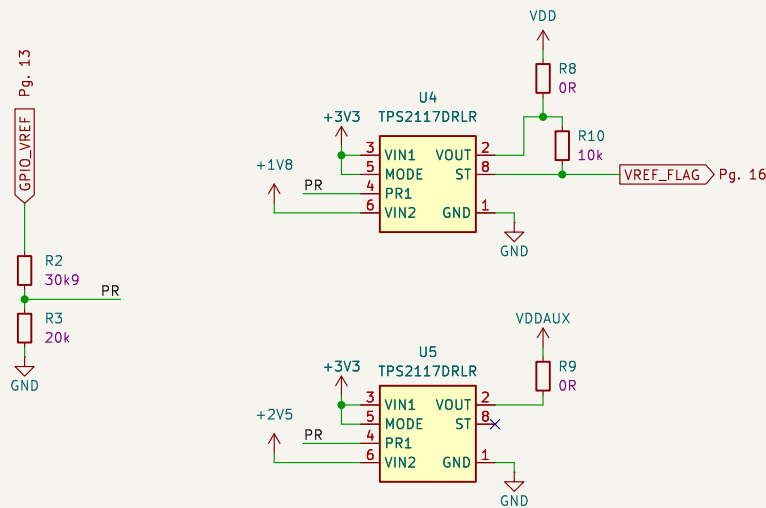
Power Monitor



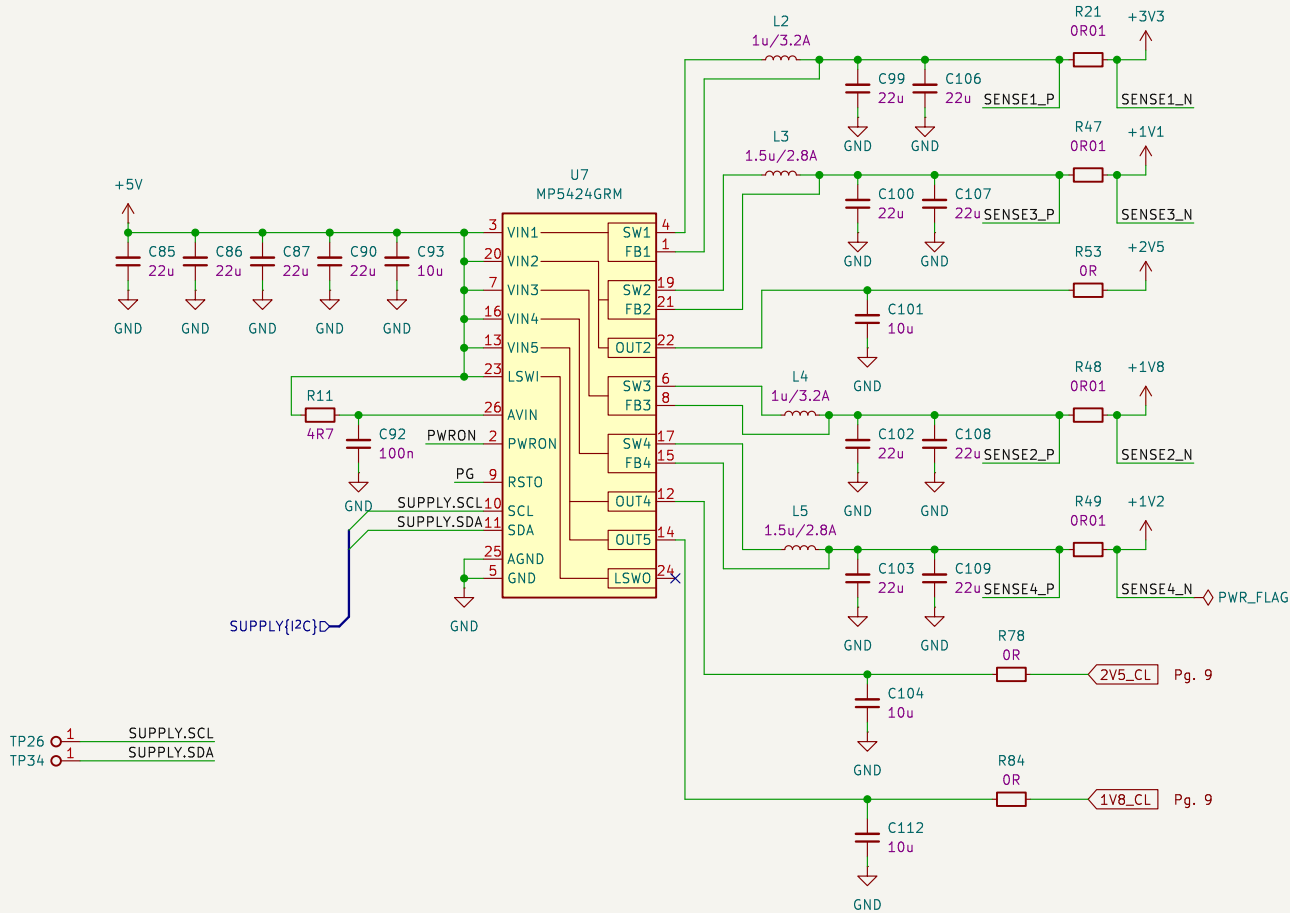
GLOBAL_EN



GPIO Power Mux

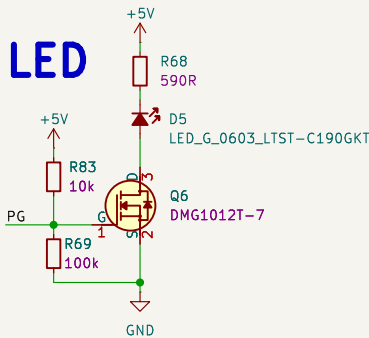


PMIC

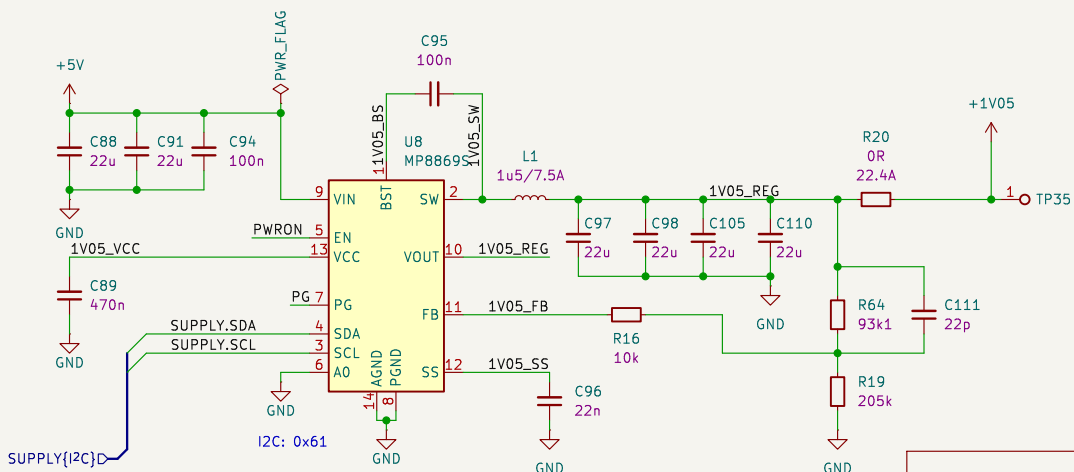


Soft start delay time slot												
	1	2	3	4	5	6	7	8	9	10	11	12
3V3			X									
1V1		X				X						
2V5				X								
1V2							X					
2V5_CL					X				X			
VDDI_CL						X						
1V8_CL							X					

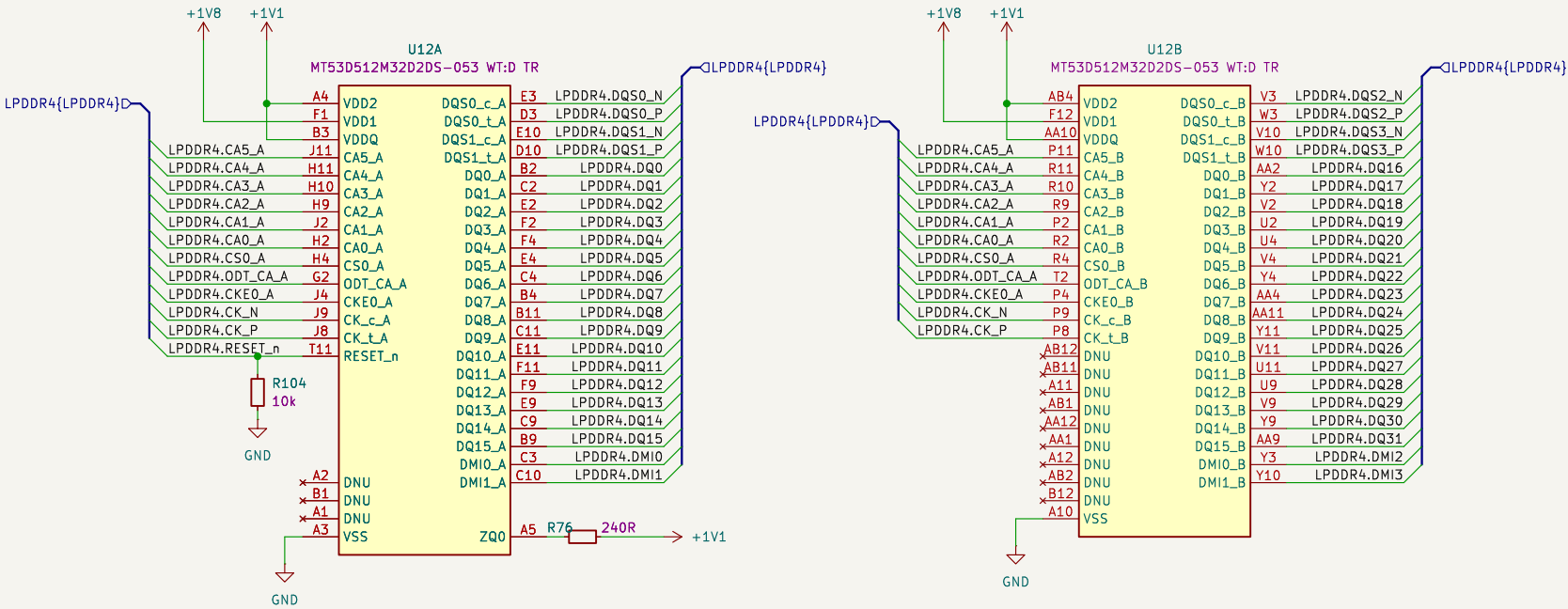
PG LED



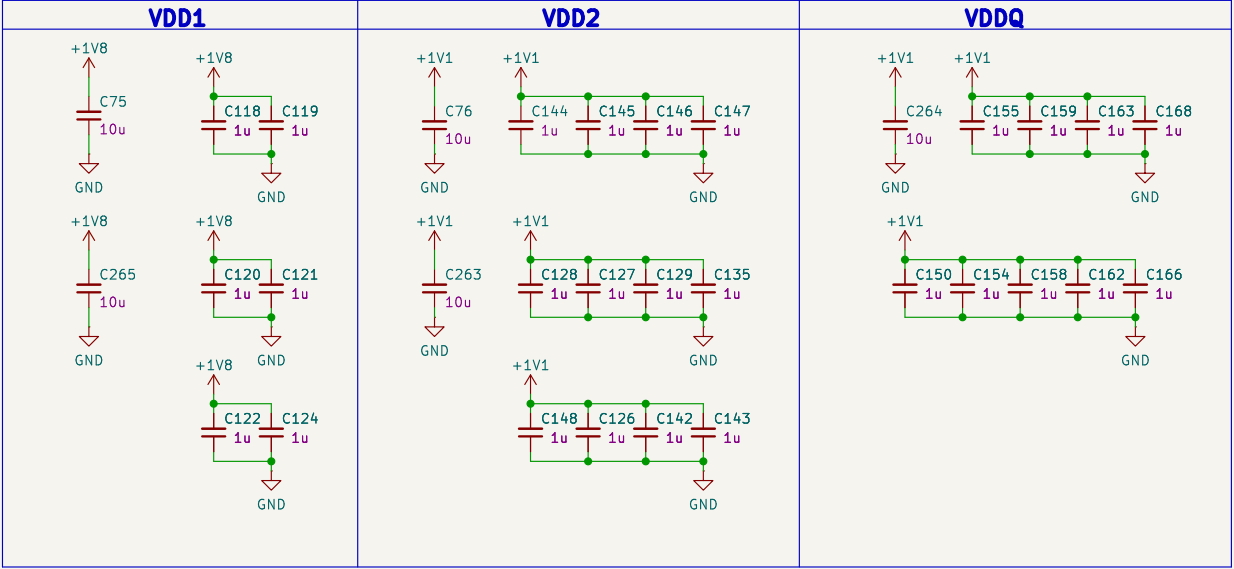
Core supply



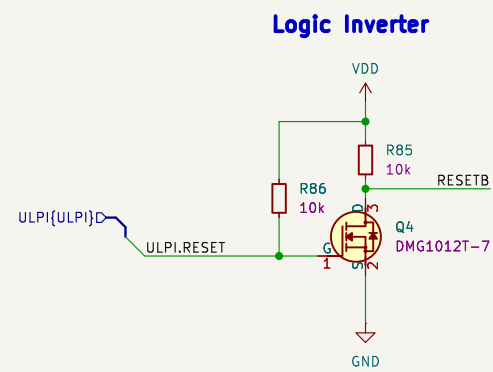
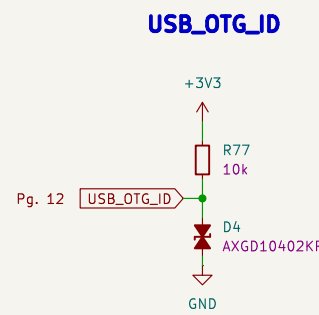
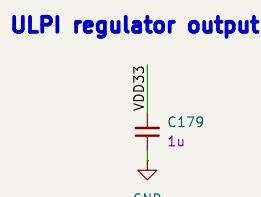
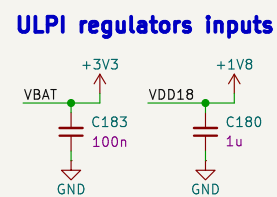
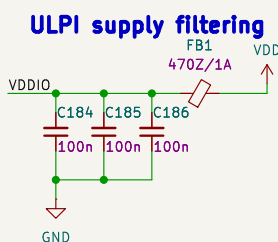
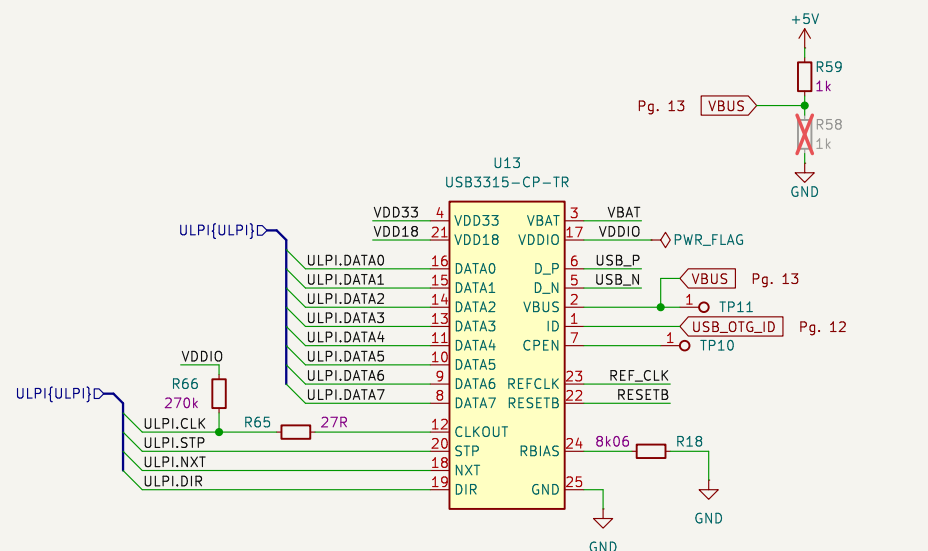
LPDDR4



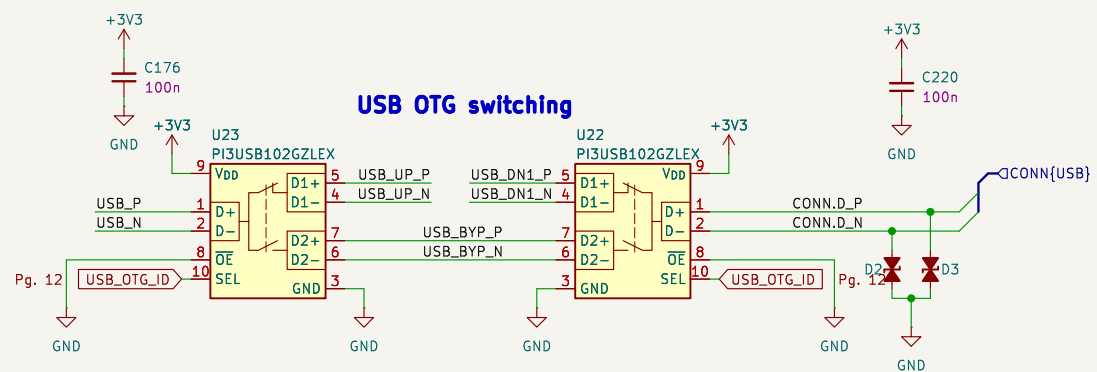
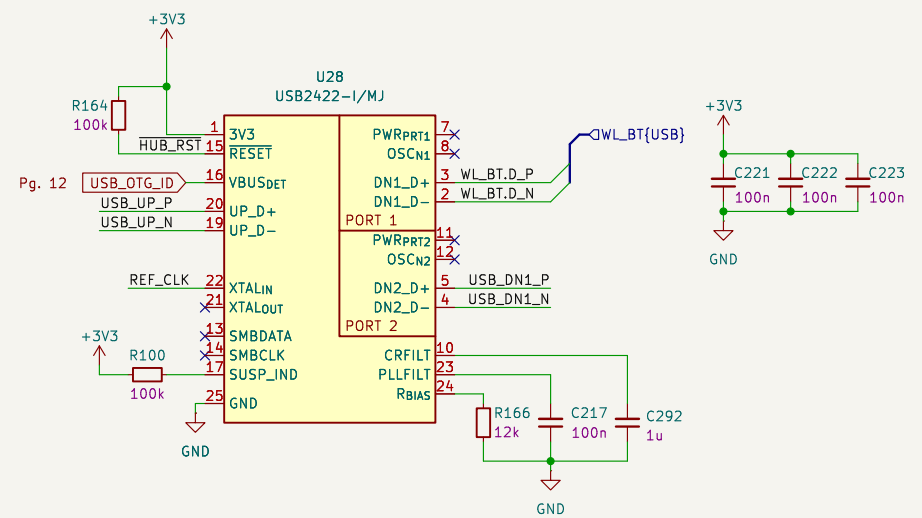
Decoupling capacitors



USB PHY



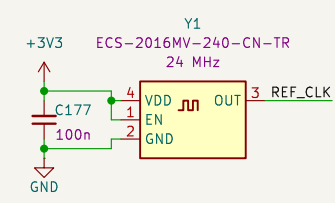
USB HUB



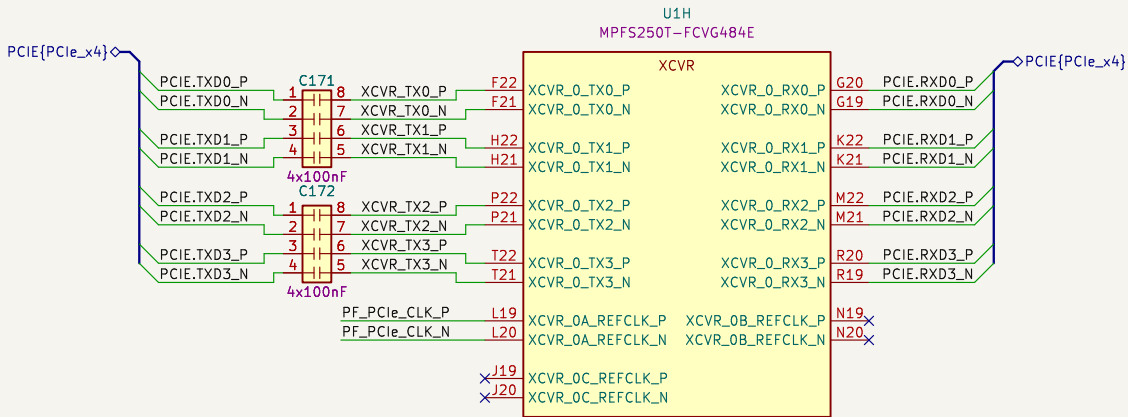
Truth table for PI3USB102G

SEL	D+	D-
L	D1+	D1-
H	D2+	D2-

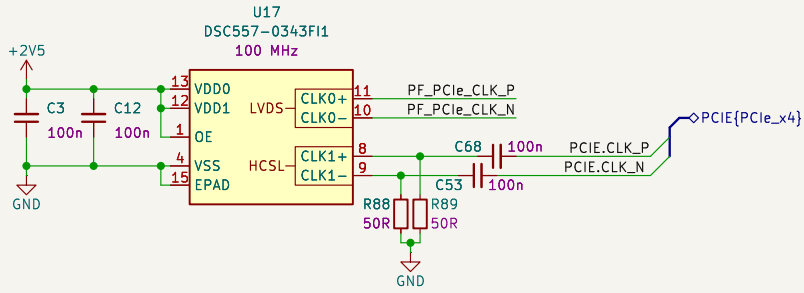
REF CLK



PolarFire XCVR

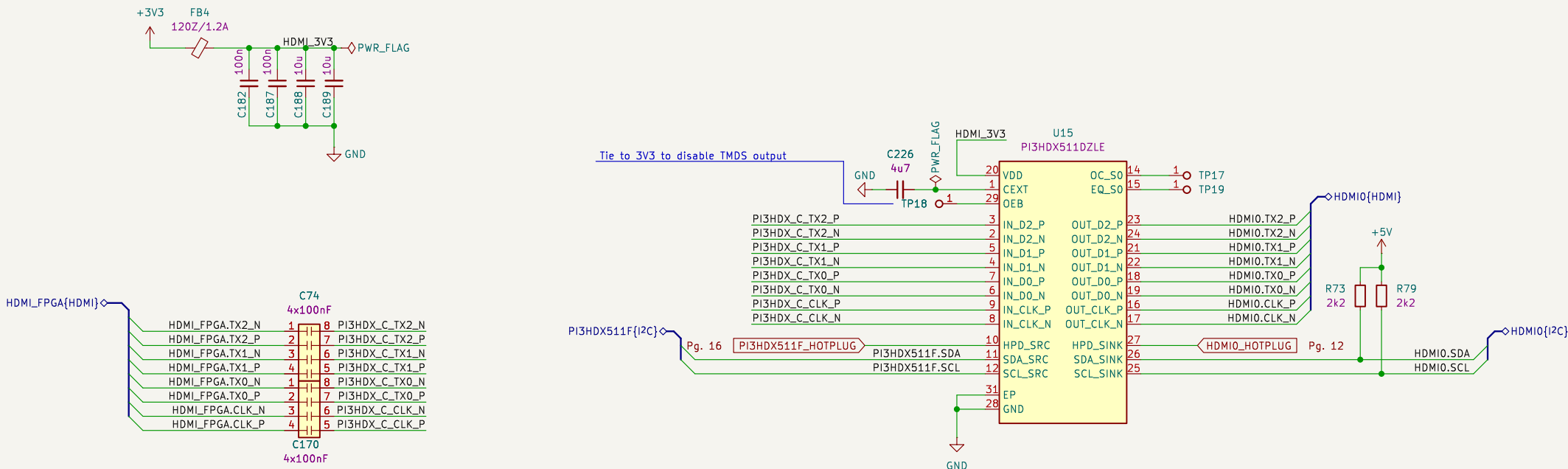


REF CLK

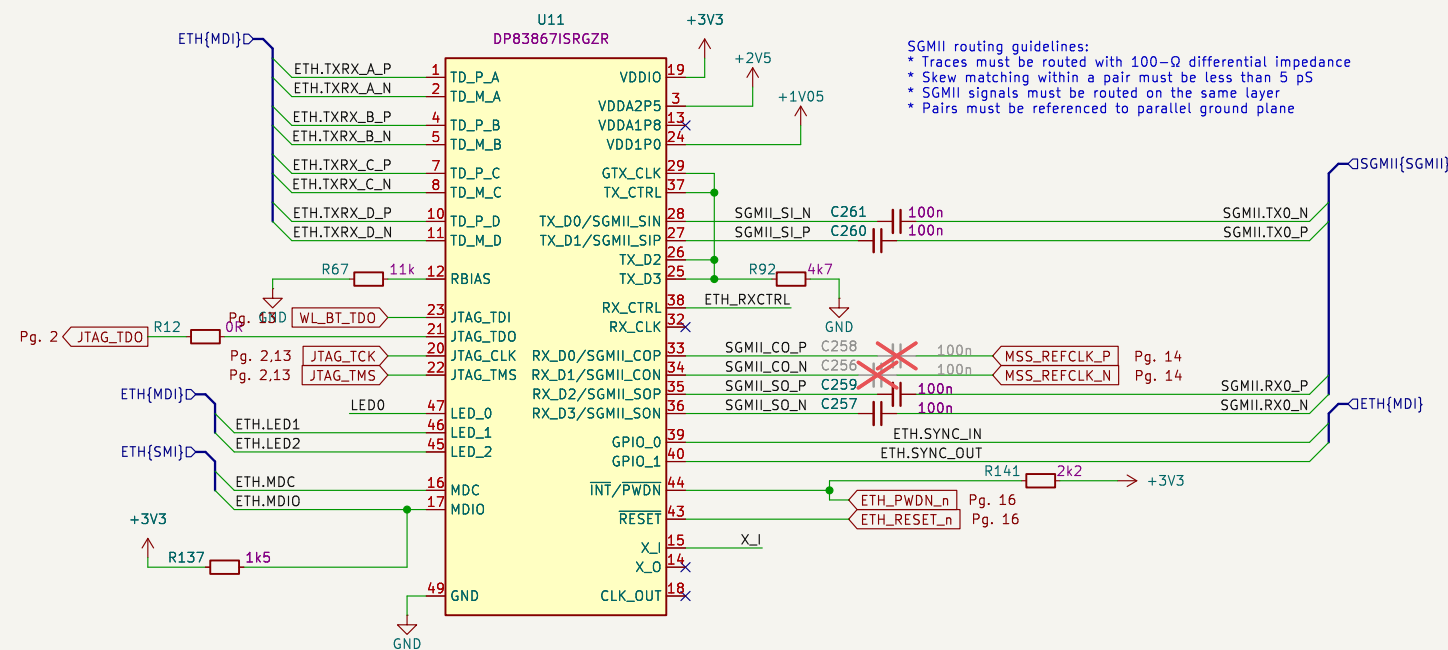




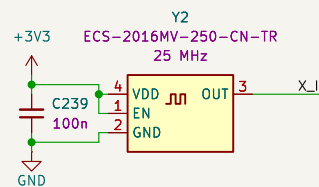
HDMI Driver



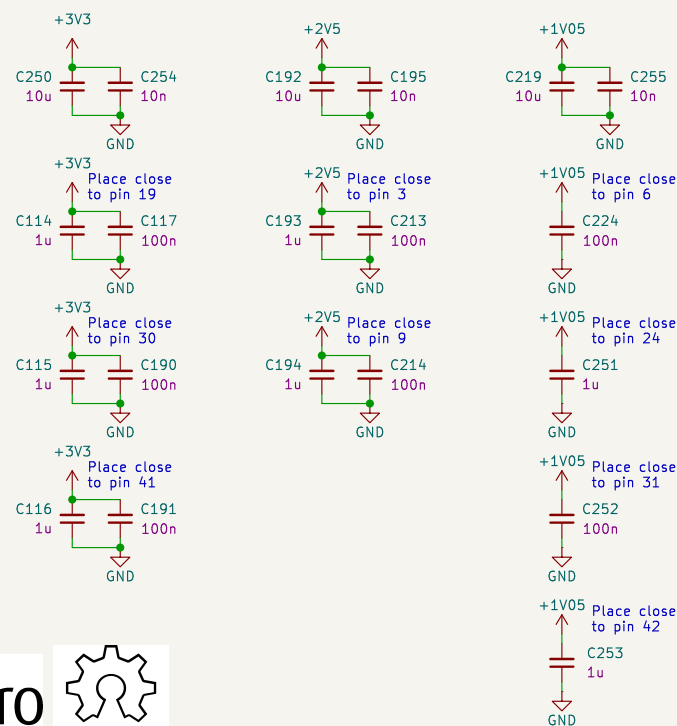
Gigabit Ethernet PHY transceiver with SGMII



REF CLK



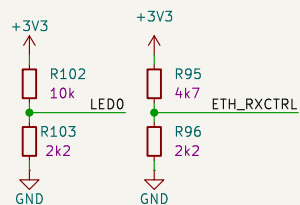
Decoupling

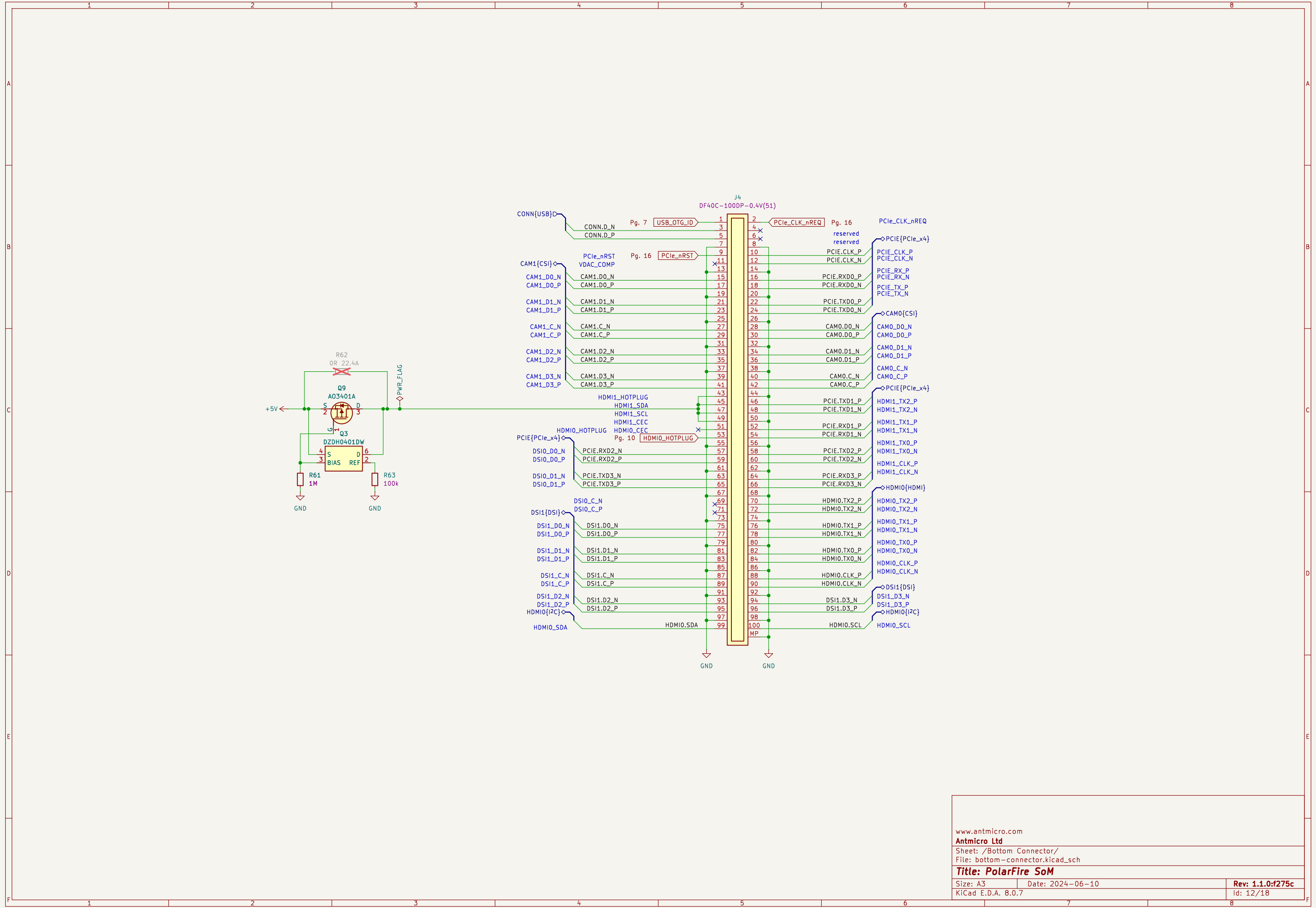


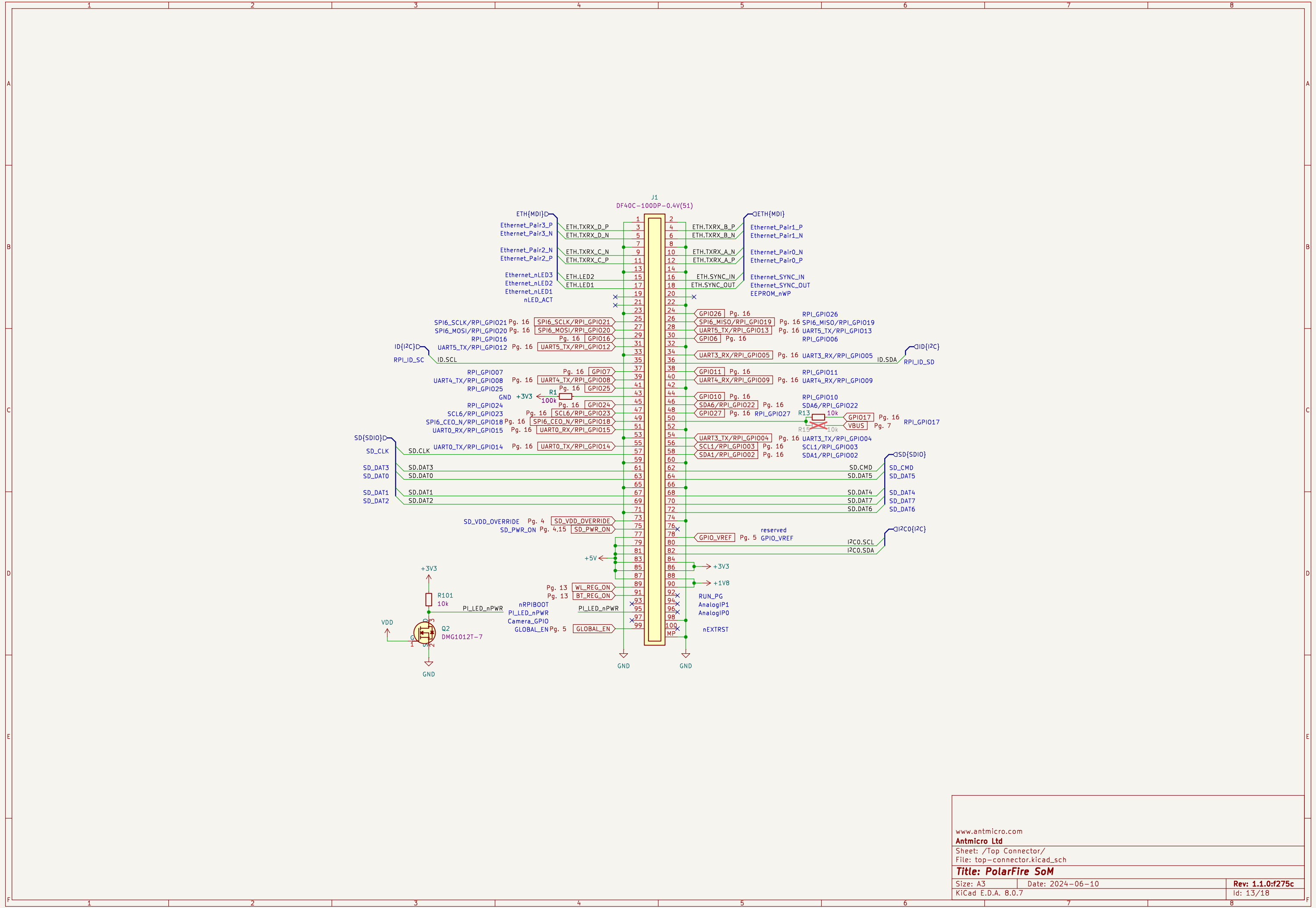
Strap resistors

Strap functions

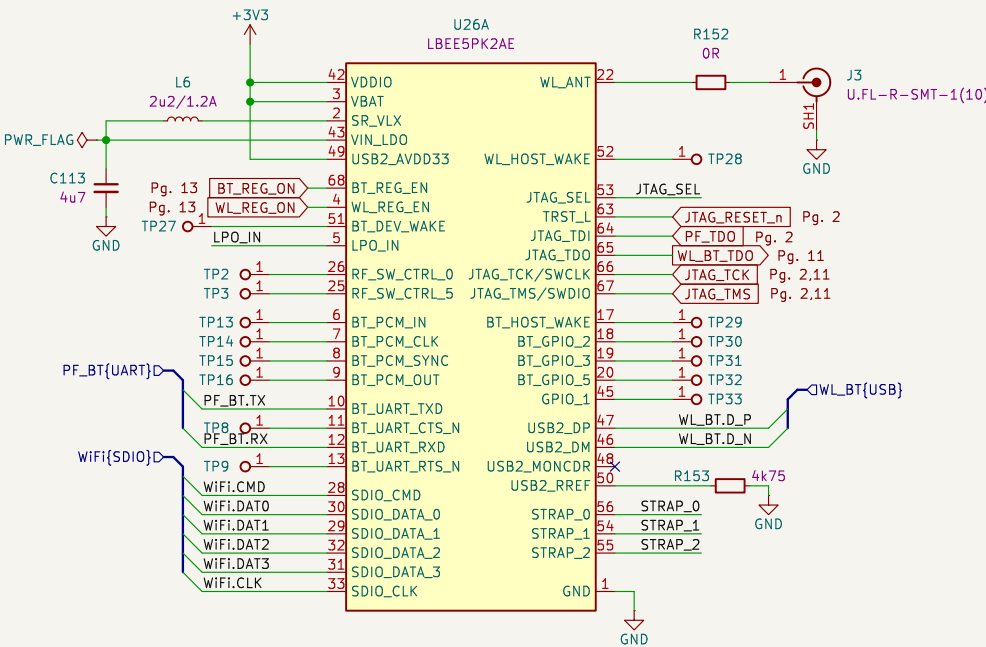
Pin name	Function	Value
RX_D0	PHY_ADD[1:0]	00
RX_D2	PHY_ADD[3:2]	00
RX_CTRL	- / Autoneg Disable	0-0
GPIO_0	RGMII Clock Skew RX[0]	0-0
GPIO_1	RGMII Clock Skew RX[2:1]	00
LED_2	RGMII Clock Skew TX[1:0]	00
LED_1	ANEG_SEL / RGMII Clock Skew TX[2]	00
LED_0	Mirror Enable / SGMII Enable	01



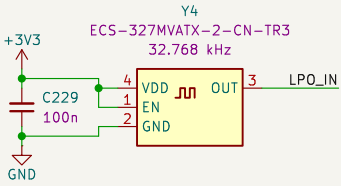




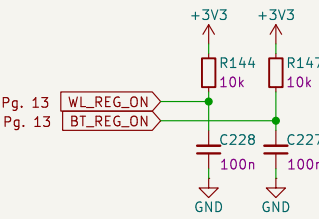
WiFi and Bluetooth Module



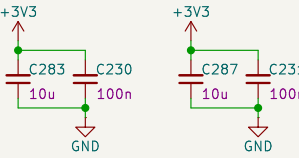
Sleep clock



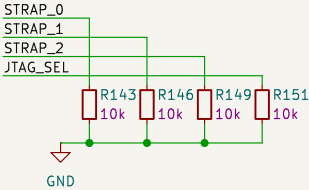
WL and BT Enable



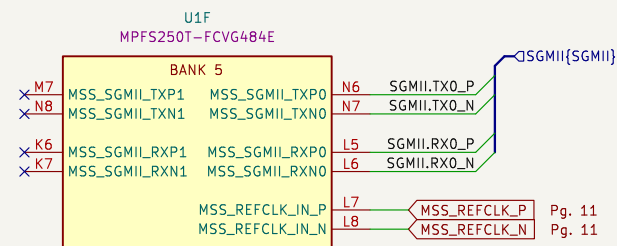
Decoupling



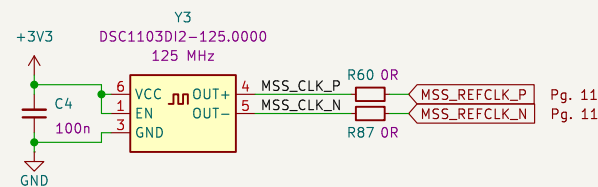
Config



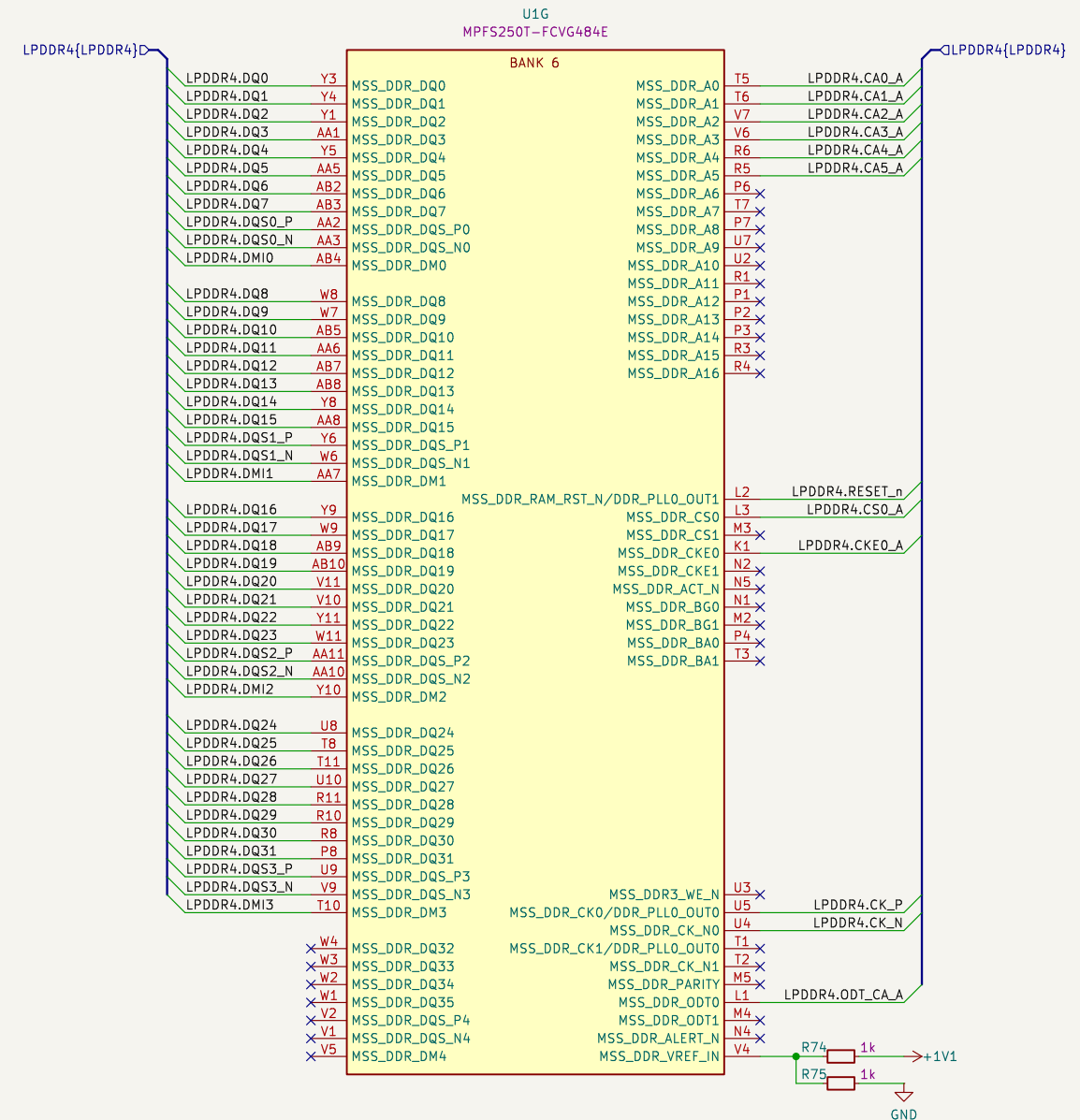
PolarFire Bank 5
MSS_SGMII



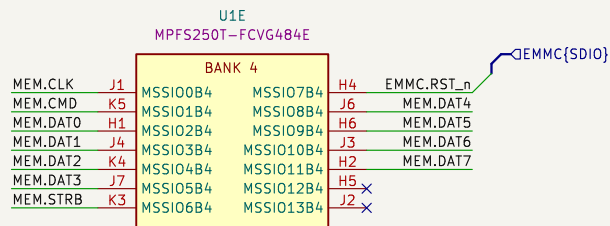
MSS CLK



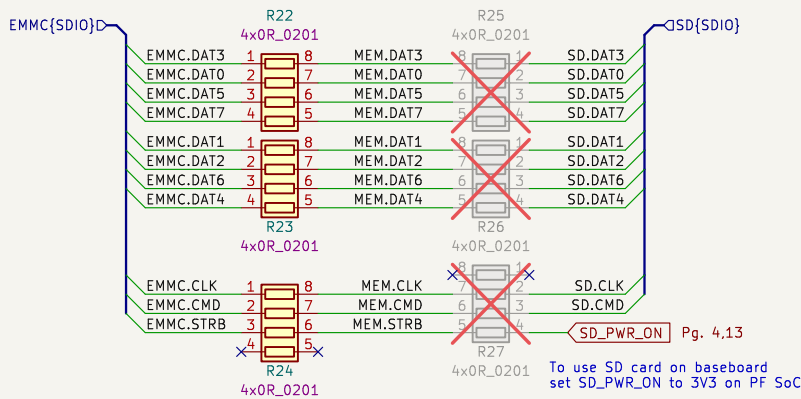
PolarFire Bank 6
MSS_DDR



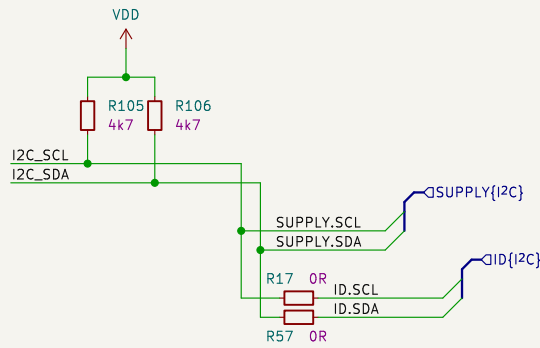
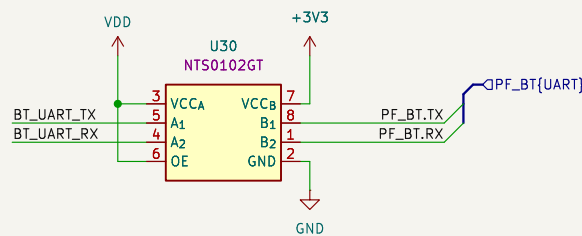
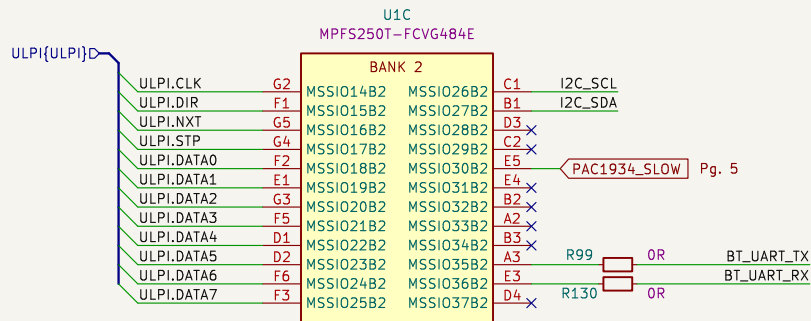
PolarFire Bank 4
MSSIO



Memory selection
eMMC SD card

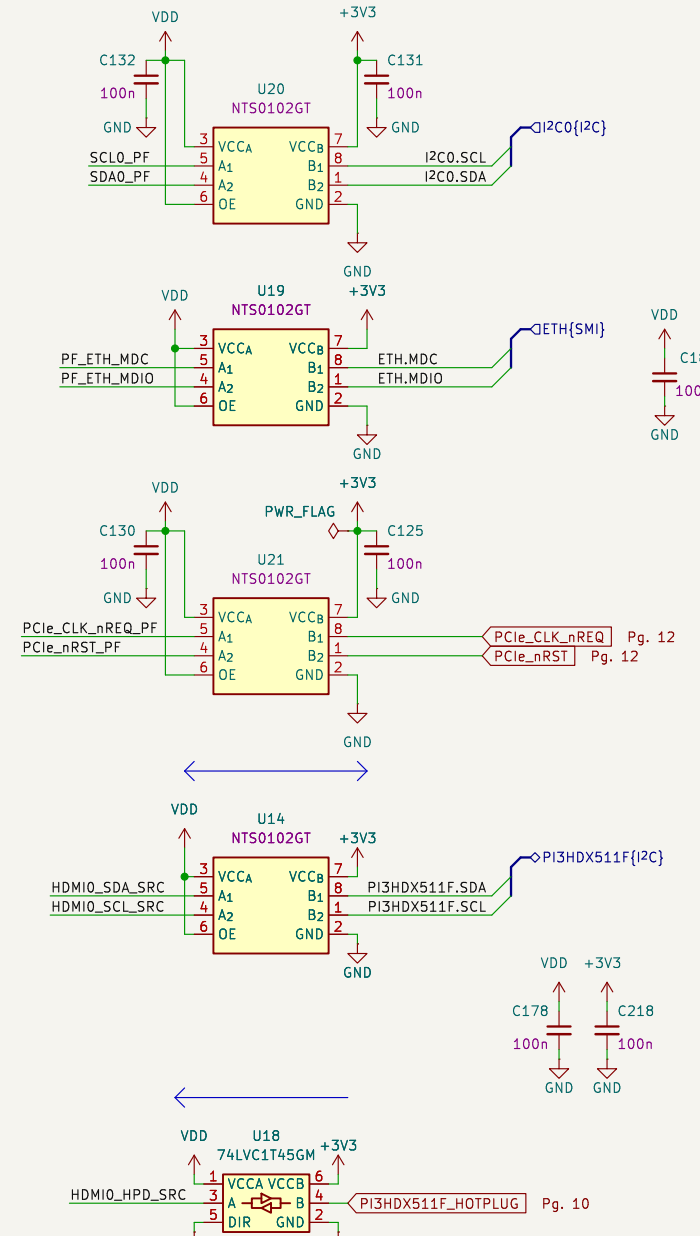
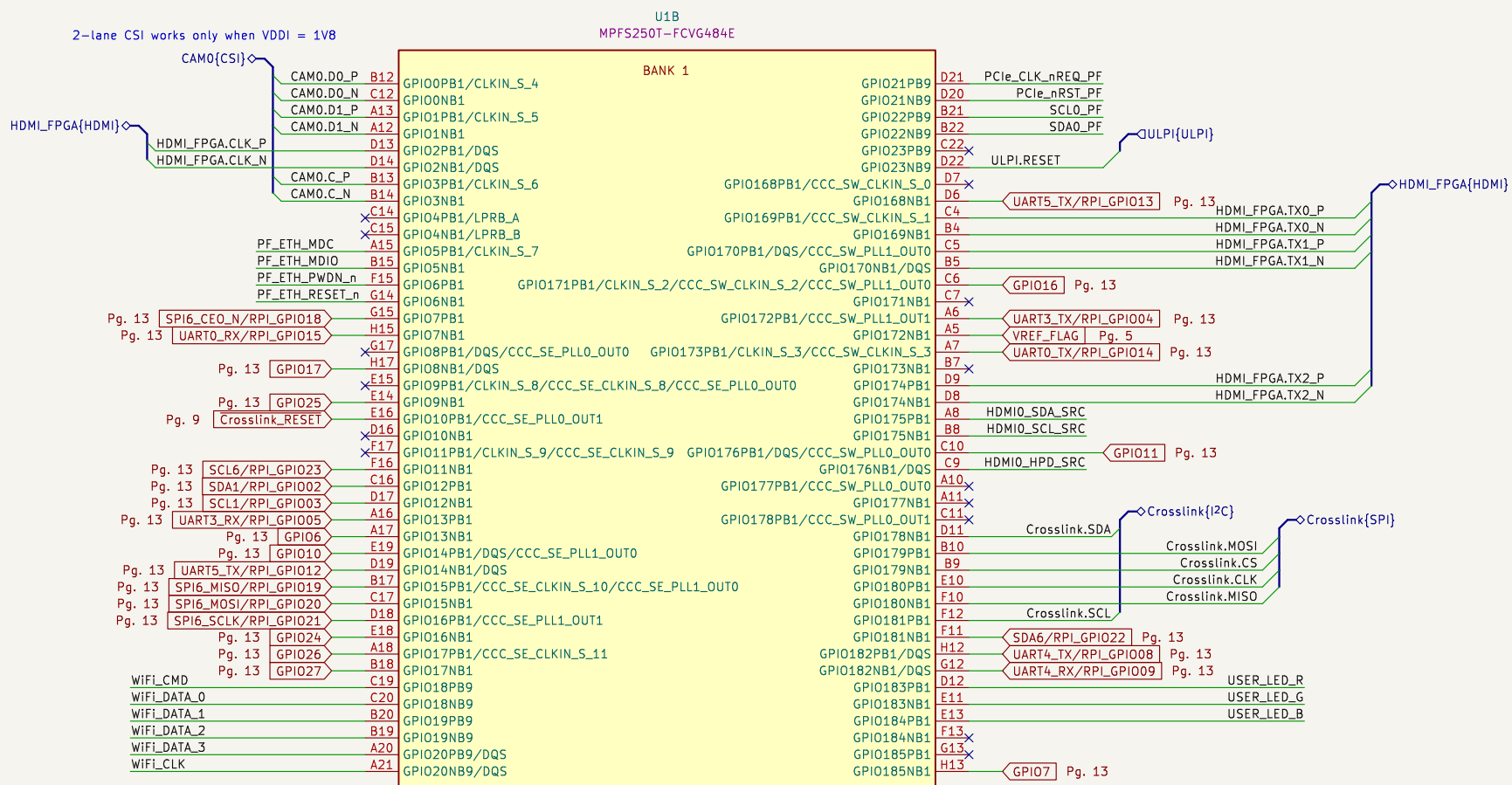


PolarFire Bank 2
MSSIO

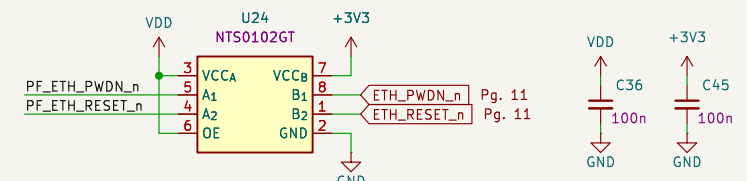
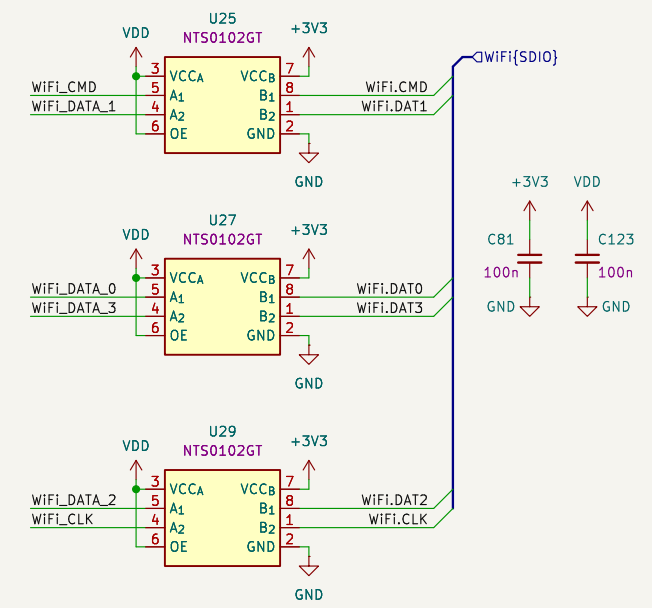
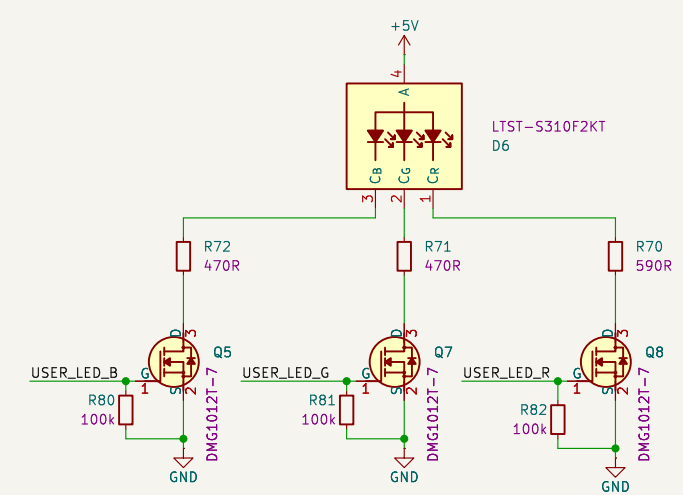


PolarFire Bank 1
GPIO

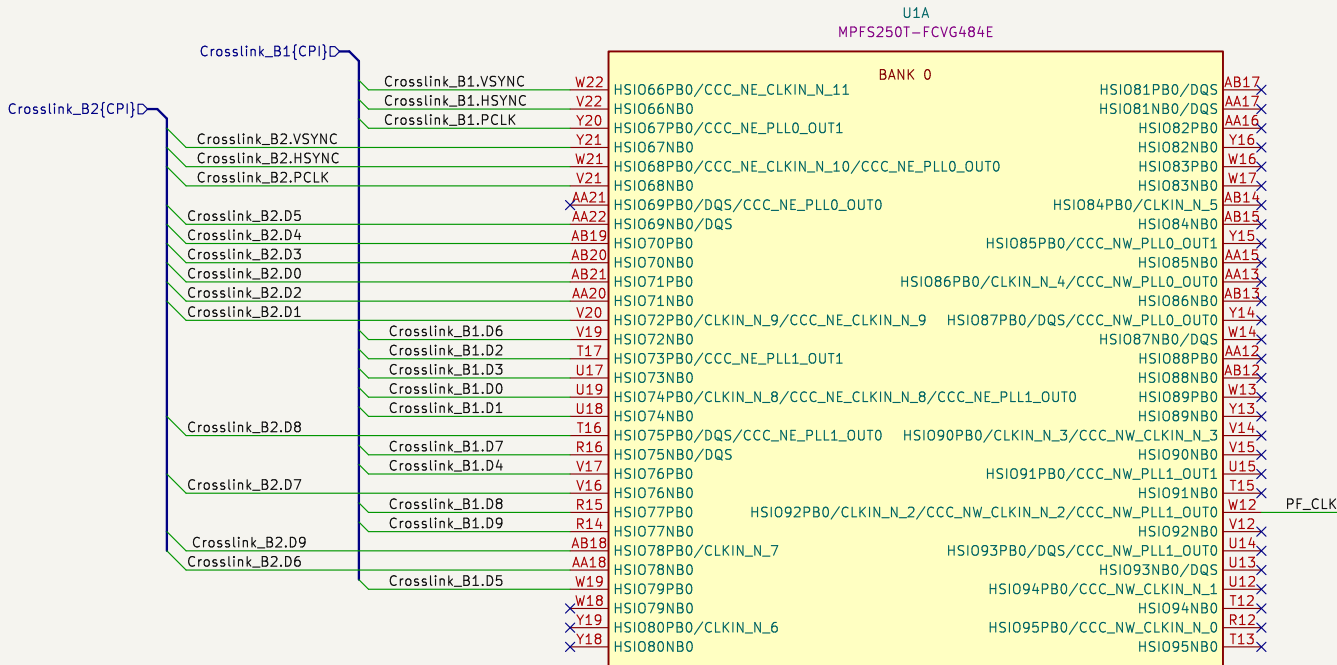
Logic shifters



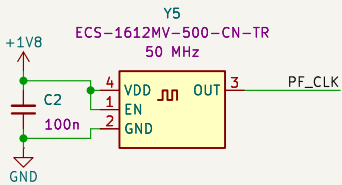
USER LED



PolarFire Bank 0 HSIO



PolarFire main clock



www.antmicro.com

Antmicro Ltd

Sheet: /FPGA HSIO/

File: fpga-hsio.kicad_sch

Title: PolarFire SoM

Size: A3

Date: 2024-06-10

Rev: 1.1.0:f275c

KiCad E.D.A. 8.0.7

Id: 17/18